

Assessment Task No.1

Doñasales Valentin Howell A

BSCS-3B

Part A – Identify the ML Problem

1. What is the application?

Game-Playing AI

2. What problem are you trying to solve?

The difficulty of the AI on each level and their level of adaptability.

3. What are the inputs?

Gameplay Videos, game plans through fan websites, and performance statistics.

4. What is the expected output?

The AI has the capability to adapt and learn throughout the whole game, such as imitating how the pro players play on a competition.

5. What patterns or relationships might the ML system learn

Pattern between player's skills, gameplay and the appropriate strategy on player.

6. What data would the system need?

Player's gameplay data, like their strategies, performance statistics, and their actions.

Part B – Traditional Programming vs Machine Learning

Traditional Programming

Input – Algorithm – Output

Rules/Instructions

1. When the player is losing, decrease the difficulty by lowering the stats
2. Vice versa, increase the difficulty by adding more stats to the ai.
3. If the player is using the same strategy each time, counter it so that the player has to come up with new strategy.
4. If the player has reached a certain level, increase the stats of the ai to be more challenging.

Machine Learning

Input + Output/Data → ML Algorithm

1. Player's performance statistics

2. Player's strategies
3. Player's Wins or Losses
4. Player's Gameplay
5. AI's performance with certain strategies
6. Difficulty of AI on a specific level

Part C – Create Your Own Mini Dataset

Input	Output
Frequently avoids Attack A	Use Attack B more often
Defeats weak enemies quickly	Increase Enemy Stats
Player's keep losing on a single level	Decrease level difficulty
Frequently gets hit by Attack B	Use Attack A more often
Survive the level for too long	Increase the difficulty by increasing the stats

Part D – Explain the Training Process

Machine Learning algorithm learns from a data through analyzing the input and output data. By identifying the relationship between the datasets, the machine can learn and adapt. The algorithm uses this pattern to make predictions on the new data it receives. Unlike the traditional programming with its fixed set of rules, the machine learning algorithm has the capable to learn and adapt.

Reflections

Why is learning from data useful instead of manually programming every possible rules?

Machine Learning has the capability to learn and adapt base on the data it receives, and the more data the it receives the better it predicts and it can decision on their own. Unlike the traditional programming, the programmer has to think of every possible rules so that it produces the desirable outcome to the user. In summary, machine learning works by learning through patterns from data to predict the output, and through training the model, the accuracy of the predictions will become more precise if it's been exposed to many datasets.